

triple integrals in cylindrical coordinates

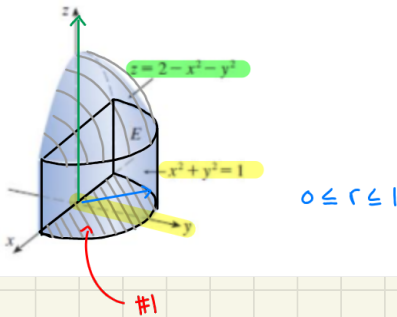
$$\iiint_E f(x,y,z) dv = \iiint f(r \cos \theta, r \sin \theta, z) r dz dr d\theta$$

for x you replace it with $r \cos \theta$ for y you replace it with $r \sin \theta$ but when it comes to z you can replace it with z

to convert from rectangular to cylindrical coordinates we use

$$r^2 = x^2 + y^2, \tan \theta = \frac{y}{x}, z = z$$

3. Express the triple integral $\iiint_E f(x,y,z) dV$ as an iterated integral in the cylindrical coordinate for the solid region E given below



#1 on a xy -Plane



if we are talking in polar coordinates it goes from 0 to π

on the y axis which we have highlighted in yellow $x^2 + y^2 = 1$ the $r = 1$

the z axis begins at zero and goes out to $z = 2 - x^2 - y^2$ which you can replace with $z = 2 - r^2$

$$\int_0^\pi \int_0^1 \int_{z=0}^{z=2-r^2} f(r \cos \theta, r \sin \theta, z) dz r dr d\theta$$

Use cylindrical coordinates to evaluate $\iiint_E \sqrt{x^2 + y^2} \, dV$ where E is the region that lies inside the cylinder $x^2 + y^2 = 16$ and between planes $z = -5$ and $z = 4$

★ the z was already given $-5 \leq z \leq 4$

$$x^2 + y^2 = 16 \Rightarrow r = \pm 4$$

• always take the positive

$$0 \leq r \leq 4$$

You are not told of any restrictions that you need to abide to so a cylinder is like a circle so you are going from $0 \leq \theta \leq 2\pi$

★ you can replace $\sqrt{x^2 + y^2}$ with r

$$\int_{\theta=0}^{2\pi} \int_{r=0}^4 \int_{z=-5}^4 r^2 \, dz \, dr \, d\theta$$

Use cylindrical coordinates to evaluate the integral $\iiint_E (x + y + z) \, dV$ where E is a solid in the First Quadrant that lies under the paraboloid $z = 4 - x^2 - y^2$



to find r you need to go into xy -Plane so you need to set $z = 0$

$$z = 4 - r^2$$

$$0 = 4 - r^2$$

$$r = \pm 2$$

• Pick Positive

★ you can rewrite z like these

$$z = 4 - r^2$$

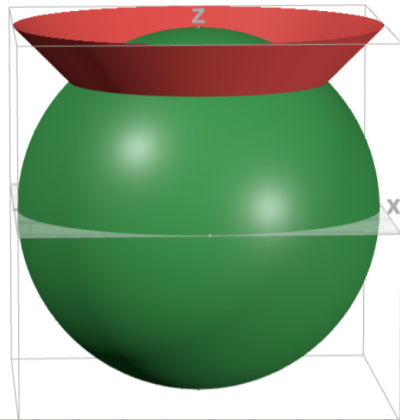
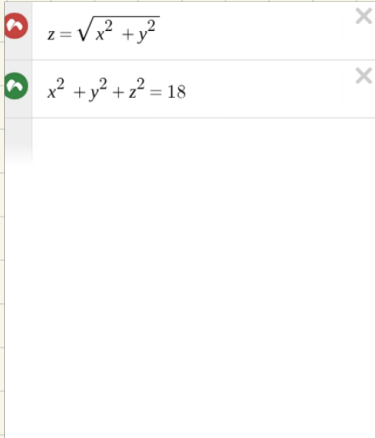
$$0 \leq z \leq 4 - r^2$$

★ it also tells you that you have a restriction which is the First Quadrant

$$0 \leq r \leq 2$$

$$\int_{\theta=0}^{\pi/2} \int_{r=0}^2 \int_{z=0}^{4-r^2} (r \cos \theta + r \sin \theta + z) \, dz \, r \, dr \, d\theta$$

Find the volume of the solid that is enclosed by the cone $z = \sqrt{x^2 + y^2}$ and the sphere $x^2 + y^2 + z^2 = 18$



★ Set the z's equal to get the intersection

$$z = \sqrt{x^2 + y^2} \Rightarrow z = r$$

$$x^2 + y^2 + z^2 = 18 \Rightarrow z = \sqrt{18 - r^2}$$

We can also see that θ is going to be going from 0 to 2π

$$0 \leq \theta \leq 2\pi$$

$$r = \sqrt{18 - r^2}$$

$$r^2 = 18 - r^2$$

$$2r^2 = 18$$

$$r^2 = 9$$

$$r = \pm 3 \quad \star \text{ always take positive}$$

$$r = 3$$

$$0 \leq r \leq 3$$

For the z-axis we will go from 0 to the top 0 to 2

$$x^2 + y^2 + z^2 = 18 \Rightarrow r^2 + z^2 = 18$$

$$z = \sqrt{18 - r^2}$$

$$0 \leq z \leq \sqrt{18 - r^2}$$

$$\int_{\theta=0}^{2\pi} \int_{r=0}^3 \int_{z=r}^{\sqrt{18-r^2}} 1 \, dz \, r \, dr \, d\theta$$

★ we are just setting it up

Find the volume of the region E that lies between the Paraboloid $z = 24 - x^2 - y^2$ and the Cone $z = 2\sqrt{x^2 + y^2}$

★ Set the two z's equal to one another to get the point of intersection

$$\begin{aligned} z &= 24 - x^2 - y^2 \rightarrow z = 24 - r^2 \\ z &= 2\sqrt{x^2 + y^2} \rightarrow z = 2r \end{aligned}$$

$$24 - r^2 = 2r$$

$$r^2 + 2r - 24 = 0$$

$$(r+6)(r-4) = 0$$

$$\cancel{r = -6} \quad r = 4 \quad 0 \leq r \leq 4$$

Pick the Positive r-value

★ There are no restrictions so θ will go from 0 to 2π , $0 \leq \theta \leq 2\pi$

it also tells you the the region E lies between $z = 24 - x^2 - y^2$ ($z = 24 - r^2$) and $z = 2\sqrt{x^2 + y^2}$ ($z = 2r$)

$$\text{So } 2r \leq z \leq 24 - r^2$$

$$\int_{\theta=0}^{2\pi} \int_{r=0}^4 \int_{z=2r}^{24-r^2} 1 \, dz \, r \, dr \, d\theta$$

When it says Find the Volume your $f(x,y,z)$ is one

evaluate the integral by changing it to cylindrical coordinates

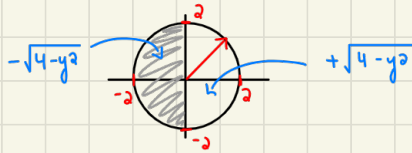
$$\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} \int_{\sqrt{x^2+y^2}}^2 xz \, dz \, dx \, dy$$

$$x = \sqrt{4-y^2}$$

$$x^2 = 4-y^2$$

$$x^2 + y^2 = 4$$

$$r = \pm 2$$



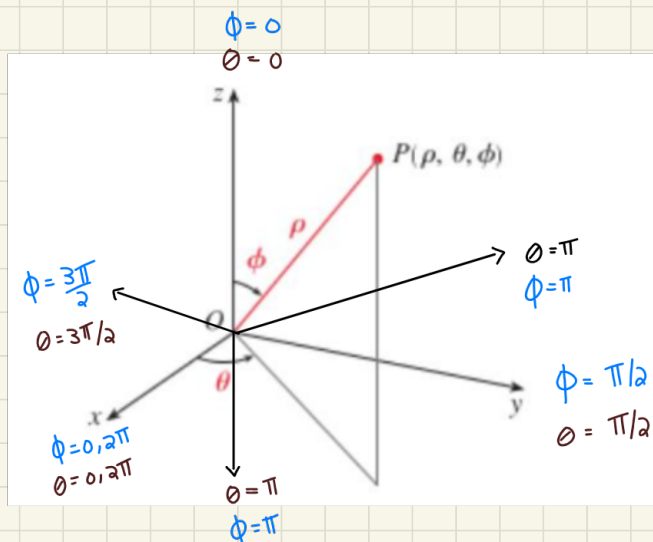
$$0 \leq \theta \leq 2\pi$$

$$0 \leq r \leq 2$$

$$r \leq z \leq 2$$

$$\int_{\theta=0}^{2\pi} \int_{r=0}^2 \int_{z=r}^2 (r \cos \theta) z \, dz \, r \, dr \, d\theta$$

• triple integral in Spherical Coordinates



$0 \leq \phi \leq \pi$ and θ is on the xy -plane and $\rho \geq 0$

equations you need to know:

★ ρ (row) goes from zero to where ever Point P is

$$z = \rho \cos \phi$$

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$r = \rho \sin \phi$$

$$\rho^2 = x^2 + y^2 + z^2$$

• Fact to know:

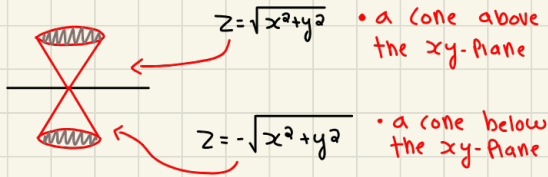
$$x^2 + y^2 + z^2 = r^2 \quad \bullet \text{ Sphere with center } (0,0,0) \text{ and radius } r$$

$$\rho = r \quad \bullet \text{ Spherical coordinate for a sphere with center } (0,0,0) \text{ and radius } r$$

remember $\rho = \sqrt{x^2 + y^2 + z^2}$

$$z^2 + x^2 + y^2$$

• Cone above and below xy-plane



$$z = \pm \sqrt{x^2 + y^2} \rightarrow \rho \cos \phi = \pm \rho \sin \phi \rightarrow \phi = \tan^{-1}\left(\frac{1}{\pm}\right)$$

if ϕ is greater than zero its above xy-plane

if ϕ is smaller than zero it below xy-plane

You will write the integral like:

$$\int_c^F \int_a^d \int_0^b f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$$

and

$$(\rho, \theta, \phi)$$

Changing rectangular coordinates into Spherical coordinates:

$$(4, 4, 0)$$

$$\rho \cos \phi = z$$

$$x = \rho \sin \phi \cos \theta$$

$$\rho^2 = (4)^2 + (4)^2 + (0)^2$$

$$\sqrt{32} \cos \phi = 0$$

$$4 = \sqrt{32} \sin(\pi/2) \cos \theta$$

$$\rho = 32$$

$$\phi = \cos^{-1}(0)$$

$$\frac{4}{\sqrt{32} \sin(\pi/2)} = \cos \theta$$

$$\rho = \sqrt{32}$$

$$\phi = \pi/2$$

$$\frac{\sqrt{2}}{2} = \cos \theta$$

Checking

$$\theta = \cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$$

You can check if this is correct taking the x and y (4,4) and plotting to see if that's where the answer you got will be on unit circle

$$\theta = \pi/4$$

$$\boxed{(\sqrt{32}, \pi/4, \pi/2)}$$

describe the surface:

$\rho=3$ • a sphere with center $(0,0,0)$ and radius 3
as $\rho = \sqrt{x^2+y^2+z^2}$

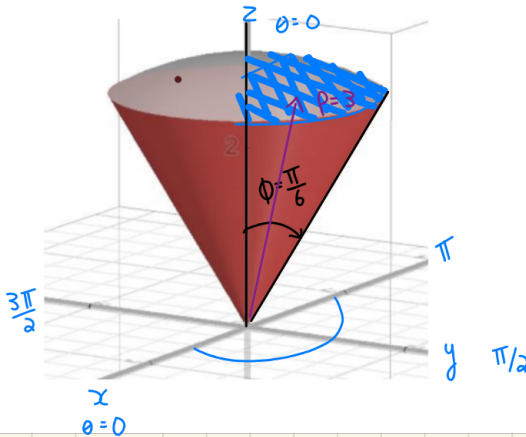
$\phi = \frac{\pi}{4}$ • equation of a cone above xy -plane
 $z = \sqrt{x^2+y^2}$

$\phi = \frac{5\pi}{6}$ • equation of plane below xy -plane
 $z = -\sqrt{x^2+y^2}$

$\rho \cos \phi = 3$ • $\rho \cos \phi = z$ so $z=3$ which is just a plane

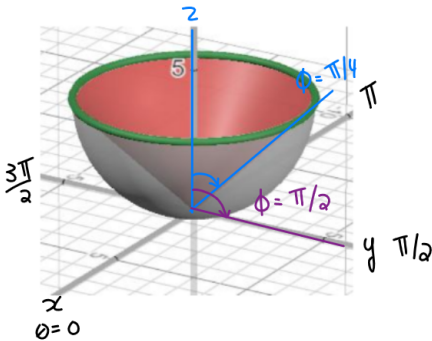
$\rho \sin \phi \cos \theta = 5$ • $\rho \sin \phi \cos \theta = x$ so $x=5$ which is just a plane

4. Sketch and describe the solid given by $\rho \leq 3, 0 \leq \phi \leq \frac{\pi}{6}, 0 \leq \theta \leq \pi$



$\rho \leq 3$ where ρ is a
sphere

5. A solid lie inside the sphere $x^2 + y^2 + z^2 = 8z$ and outside the cone $z = \sqrt{x^2 + y^2}$. Write the solid in terms of the inequalities involving spherical coordinates.



You need to use the cone equation to find ϕ

$$z = \sqrt{x^2 + y^2}$$

$$P \cos \phi = P \sin \phi$$

$$1 = \frac{\sin \phi}{\cos \phi}$$

$$1 = \tan \phi$$

$$\phi = \tan^{-1}(1)$$

$$\phi = \pi/4$$

$$x^2 + y^2 + z^2 = 8z$$

$$\rho^2 = 8\rho \cos \phi$$

$$\rho = 8 \cos \phi$$

$$0 \leq \rho \leq 8 \cos \phi$$

$$0 \leq \rho \leq 8 \cos \phi$$

$$\pi/4 \leq \phi \leq \pi/2$$

$$0 \leq \theta \leq 2\pi$$

0 would go all around it as its circular
So:

$$0 \leq \theta \leq 2\pi$$

this $\phi = \pi/4$ is for the cone

this $\phi = \pi/2$ is for the sphere

$$\pi/4 \leq \phi \leq \pi/2$$

You can also find the radius of the sphere

$$x^2 + y^2 + z^2 = 8z \quad \cdot \text{apply}$$

$$x^2 + y^2 + z^2 - 8z = 0$$

$$\downarrow \div 2$$

$$(x^2 + y^2 + (z-4)^2) = 16$$

$$x^2 + y^2 + z^2 - 8z + 16 = 16$$

$$x^2 + y^2 + (z-4)^2 = 16$$

$$r = 4$$